Group Topic: EduPulse | Lab No. WEEK#2

Git Repo/ Colab Link: https://github.com/Drexgithub/Group-Repository | Date:04/23/2026

Objectives

The task's goal is to obtain a dataset of at least 500,000 rows from a single file while closely following procurement and data ethics guidelines. In order to guarantee that the final dataset is precise, polished, and prepared for analysis, the emphasis switched to thorough data processing and cleaning after the data collection stage.

Tasks Completed

1. Data Procurement

To meet the project's data procurement objectives, the Student Performance dataset—originally consisting of 1,000,000 rows within a structured CSV file—was sourced and subsequently trimmed to exactly 500,000 rows to align with task specifications. During the pre-processing phase, a focus on data ethics led to the removal of sensitive proxy variables, specifically the attendance_percentage and class_participation columns. The final refined dataset was saved as student_performance_clean.csv, featuring four primary columns: student_id, weekly_self_study_hours, total_score, and grade.

2. Load Dataset into a Spark DataFrame

The cleaned CSV was loaded into a Spark DataFrame using PySpark. The schema was inferred automatically, and the data was partitioned across executors for distributed processing.

```
from pyspark.sql import SparkSession
 spark = SparkSession.builder.appName('StudentPerformance').getOrCreate()
 df = spark.read.csv('student_performance_clean.csv', header=True,
inferSchema=True)
 df.printSchema()
 df.show(5)
```

3. Exploratory Data Analysis (EDA)

```
>>> # Check missing/null values per column
>>> print("=== Missing Values ===")
=== Missing Values ===
>>> df.select([
...     F.count(F.when(F.col(c).isNull(), c)).alias(c)
...     for c in df.columns
... ]).show()
+----------+----------------------+-----------+-----+
|student_id|weekly_self_study_hours|total_score|grade|
+----------+----------------------+-----------+-----+
|         0|                     0|          0|    0|
+----------+----------------------+-----------+-----+
```

The Exploratory Data Analysis (EDA) evaluated the dataset's structure and distribution , confirming that all 500,000 rows were complete with zero missing values. Using the Interquartile Range (IQR) method for outlier detection and logical filtering , the analysis validated two primary numeric features: weekly self-study hours and total score. This process ensured high data quality, revealing that all records fell within logical bounds, such as scores not exceeding 100. Final summary statistics underscored the dataset's reliability, notably highlighting that weekly study hours are a powerful predictor of academic success.

4. Filtering Invalid Records

Filtering rules were applied to remove logically invalid records from the dataset:

- Removed records where total_score > 100 (impossible score) — 0 records found.
- Removed records where weekly_self_study_hours < 0 (negative hours) — 0 records found.

Result: All 500,000 records passed validation. The dataset is clean and logically consistent.

```
df_filtered = df.filter((df.total_score <= 100) & (df.weekly_self_study_hours >= 0))
 print(f'Records after filtering: {df_filtered.count()}')
```

Tasks to be Completed

- Create data visualizations (histograms, scatter plots, box plots) using matplotlib/seaborn to illustrate grade distribution and outlier spread.
- Perform time-series analysis if timestamped data becomes available in future dataset iterations.

Results and Analysis

Data Quality

```
+----------+-----------------------+------------+-----+
|student_id|weekly_self_study_hours|total_score|grade|
+----------+-----------------------+------------+-----+
|         0|                      0|           0|    0|
+----------+-----------------------+------------+-----+
```

Post-filtering results revealed that the dataset contained no missing entries or logically inconsistent records. This high level of data integrity confirms that the dataset is both robust and reliable for future stages of the project, including advanced analytical processing and the training of machine learning models.

Grade Distribution

```
>>>
>>> grade_dist.show()
+-----+------+----------+
|grade| count|percentage|
+-----+------+----------+
|    A|274151|     54.83|
|    B|129399|     25.88|
|    C| 70731|     14.15|
|    D| 22571|      4.51|
|    F|  3148|      0.63|
+-----+------+----------+
```

With over half of the students (54.83%) achieving an A grade, the dataset indicates a high-achieving demographic. This results in a right-skewed distribution, where the volume of students decreases significantly in the lower grade categories.

Challenges and Solutions

Challenge: The original file size 1,000,000 rows exceeded the project requirement.
Solution: Employed pandas head(500000) to precisely trim the dataset before Spark ingestion.

Challenge: Presence of ethically sensitive proxy variables (attendance/participation).

Solution: Dropped these columns during pre-processing to comply with ethics standards.

Challenge: Local memory constraints during initial inspection of the full dataset.

Solution: Leveraged Spark's distributed architecture and pandas row sampling for efficient handling.

Conclusion

This lab successfully validated a 500,000-row dataset for distributed analysis. Key findings show that **weekly self-study hours** serve as a massive predictor of success—A-grade students averaged over **13x more study time** than those receiving an F. By utilizing Apache Spark, the workflow proved that large-scale data can be processed efficiently while maintaining high standards of data quality. The pre-processing phase proved vital in this big data workflow, as it allowed for the removal of sensitive variables that could have introduced bias into future predictive models. Furthermore, the exploratory analysis confirmed that the data is complete and logically sound, providing a reliable foundation for upcoming tasks, such as creating advanced visualizations and potentially performing time-series analysis. Final results underscore the reliability of using distributed computation to handle large datasets without compromising on accuracy or ethical standards.